

G3 Eleven-Week Roadmap

AMS Intake Platform - Algorithmic Intelligence Engine

Prepared by: Hugo Davion & Axel Brazeau
Group 3 — Algorithmic Intelligence Engine
<https://github.com/hodncgi/AMSSchatBot>

1. Introduction

This document describes the week-by-week delivery plan for Group 3. Weeks 1–8 deliver the core POC (chat-first from Week 3, algorithms Week 5, G2 API alignment Week 6, feature-complete by Week 8). Weeks 9–11 are reserved for manager feedback, Group 2 integration testing, and cross-project adaptations — not new core features.

2. Eleven-Week Plan

Structure: Weeks 1–8 = core technical delivery · Weeks 9–11 = stabilization buffer (manager feedback, G2 live integration, adaptations). Current position: between Week 2 and 3.

Week 1 - Project launch and user needs

Goals:

- Define Group 3 scope with referent and Group 2.
- Identify user personas: new employee, intake consultant, manager.
- Draft user stories and high-level requirements.

Tasks:

- Kickoff meeting and role allocation (Hugo and Axel).
- Map onboarding page as sole chatbot entry point.
- List questions users must be able to ask (onboarding and risk analysis).
- Write first version of user stories (US-01 to US-16).

Deliverables:

- User stories v0.1
- Persona list
- Scope document

Linked user stories: US-01, US-11

Week 2 - Architecture and API design

Goals:

- Design backend architecture for NBQ, Change Risk, and Chat.
- Choose local AI stack (LM Studio + Mistral).
- Define how the chatbot uses all G3 backend routes.

Tasks:

- Architecture diagram: Onboarding UI to FastAPI to NBQ, Change Risk, Chat, Health.
- API contract draft for Group 2.

- Document chatbot knowledge scope (RAG and API-aware answers).
- LM Studio installation and model selection.

Deliverables:

- Architecture note
- API contract draft
- LM Studio ready

Linked user stories: US-08, US-12

Week 3 - Backend foundation and minimal chat

Goals:

- Create FastAPI project structure.
- Expose GET /health and first POST /chat (Mistral direct, no RAG).

Tasks:

- backend/ setup: main.py, config.py, ai_service.py.
- GET /health (API, LM Studio status).
- POST /chat — LM Studio only, simple system prompt.
- CORS for onboarding frontend (localhost:3000).

Deliverables:

- FastAPI on port 8000
- GET /health
- POST /chat (minimal)

Linked user stories: US-01, US-08

Week 4 - Usable chat UI

Goals:

- Wire a simple chat interface to the live API.
- Enable suggested starter questions for demos.

Tasks:

- HTML/JS chat page (Figma-inspired shell) calling POST /chat.
- Suggested questions UI (US-02).
- Display assistant replies and basic AI mode indicator.
- README and SETUP run instructions for end-to-end demo.

Deliverables:

- Live HTML chat page
- Suggested questions
- End-to-end chat demo

Linked user stories: US-01, US-02, US-07

Week 5 - NBQ and Change Risk algorithms

Goals:

- Deliver both deterministic algorithms in one week.
- Document decision-support scope for change risk.

Tasks:

- POST /nbq/next — weighting rules, Mistral why field.
- POST /change-risk — drivers, edge cases (score min 0).
- pytest for both endpoints.
- Algorithm concepts in chat knowledge (pre-RAG).

Deliverables:

- POST /nbq/next
- POST /change-risk
- Algorithm tests

Linked user stories: US-03, US-04, US-05, US-06, US-11

Week 6 - Group 2 API alignment

Goals:

- Receive API routes from the collaborating G2 team (contact pending).
- Map G2 routes to G3 endpoints and update integration docs.

Tasks:

- G2 handover session when contact is established.
- Document G2 routes in G2_INTEGRATION.md.
- Align JSON contracts; log change requests.
- Configure CORS for G2 base URL when known.
- Draft joint integration test plan (Weeks 9–10).

Deliverables:

- G2 ↔ G3 route mapping table
- Updated G2_INTEGRATION.md

- Integration test plan

Linked user stories: US-08

Week 7 - Knowledge base, RAG and chat quality

Goals:

- Connect RAG to POST /chat.
- Meet sources, fallback, and uncertainty acceptance criteria.

Tasks:

- knowledge_base.py + catalog.json (~35 chunks).
- rag_service.py keyword retrieval.
- algorithm_context.py for live NBQ/risk in chat.
- Sources (US-13), uncertainty (US-14), fallback (US-09).

Deliverables:

- RAG-backed /chat with sources
- Fallback mode
- US-12 single-assistant coverage

Linked user stories: US-01, US-09, US-12, US-13, US-14, US-15

Week 8 - React UI and core POC complete

Goals:

- Ship Figma → React onboarding chatbot.
- Deliver G3 test bench and pytest — feature-complete POC.

Tasks:

- ChatInterface on Onboarding tab, Vite proxy.
- AiPipelineBar, suggested questions.
- frontend/g3-test HTML bench.
- pytest suite, start.ps1, API v1.0 documentation.

Deliverables:

- Chatbot on port 3000
- GET /g3-test
- Core POC complete at end of Week 8

Linked user stories: US-01, US-02, US-07, US-10, US-13

Week 9 - Manager feedback and UX polish

Goals:

- Incorporate manager critiques and refinement requests.
- Polish chatbot UX without new core features.

Tasks:

- Manager demo and feedback session.
- UI/UX fixes on Onboarding tab.
- Documentation updates from review comments.
- Bug fixes from internal testing.

Deliverables:

- Feedback change log
- Polished UI
- Updated docs

Linked user stories: US-07, US-14

Week 10 - G2 integration and cross-project adaptation

Goals:

- Run live G2 ↔ G3 integration tests with received API routes.
- Adapt to new manager requests or related project needs.

Tasks:

- Joint integration test with G2 team.
- CORS, auth, and timeout tuning for G2 environment.
- Handle scope adjustments from manager or parallel projects.
- Extended pytest scenarios for integration paths.

Deliverables:

- G2 integration test report
- Stabilized API v1.0

Linked user stories: US-08, US-09, US-10

Week 11 - Final delivery

Goals:

- Package project for handover.
- Present final demo and v2 roadmap.

Tasks:

- Repository cleanup and GitHub sync.
- Full demo script: chat, NBQ, change risk.
- Validate user stories US-01 to US-16.
- v2 roadmap: G1 database (US-15, US-16), semantic RAG, Docker.

Deliverables:

- Manager presentation
- Final deliverables pack
- v2 roadmap document

Linked user stories: US-01, US-05, US-06, US-12, US-15, US-16

3. Roadmap Overview

Week	Theme	Key deliverable	User stories
1	Project launch and user needs	User stories v0.1	US-01, US-11
2	Architecture and API design	Architecture note	US-08, US-12
3	Backend foundation and minimal chat	FastAPI on port 8000	US-01, US-08
4	Usable chat UI	Live HTML chat page	US-01, US-02, US-07
5	NBQ and Change Risk algorithms	POST /nbq/next	US-03, US-04, US-05, US-06, US-11
6	Group 2 API alignment	G2 ↔ G3 route mapping table	US-08
7	Knowledge base, RAG and chat quality	RAG-backed /chat with sources	US-01, US-09, US-12, US-13, US-14, US-15
8	React UI and core POC complete	Chatbot on port 3000	US-01, US-02, US-07, US-10, US-13
9	Manager feedback and UX polish	Feedback change log	US-07, US-14
10	G2 integration and cross-project adaptation	G2 integration test report	US-08, US-09, US-10
11	Final delivery	Manager presentation	US-01, US-05, US-06, US-12, US-15, US-16

4. Backend Routes

Route	Purpose	Used by
POST /chat	Onboarding conversation (Mistral + RAG)	Onboarding chatbot

POST /change-risk	Change risk score 0-100 + drivers	Test bench, chatbot knowledge
POST /nbq/next	Next best intake question + why	Test bench, chatbot knowledge
GET /health	API, LM Studio, RAG status	Chatbot UI, operators

5. Conclusion

This roadmap compresses core delivery into eight weeks with a chat-first approach. Week 6 schedules Group 2 API route intake (collaborating team not yet in contact). Weeks 9–11 absorb manager feedback, integration testing, and scope adjustments. Week 11 closes with a management-ready demo and validated user stories. Version 2 adds Group 1 database integration (US-15, US-16) and semantic RAG.

Hugo Davion & Axel Brazeau
Group 3 — Algorithmic Intelligence Engine